

Dark-photon production by Compton scattering

A derivation and a test of Park's Figure 1

Independent ROOT/C++ calculation

17 September 2026

Abstract

We derive the unpolarized tree-level cross section for a real photon scattering on an electron at rest to produce a massive dark photon. The derivation starts with the two electron-exchange diagrams. An exact polynomial trace calculation and independent Dirac matrices check the result. We fold the cross section with Park's reactor spectrum and compare against coordinates extracted afresh from the supplied PDF. The calculation passes the cross-section checks but does not reproduce the full published figure. No later correction is included.

1 Question, assumptions and background formulas

The process and notation are

$$\gamma(k) + e^-(p) \longrightarrow A'(q) + e^-(p'), \quad m = m_e, \quad M = m_{A'}. \quad (1)$$

We use the metric and interaction

$$g_{\mu\nu} = \text{diag}(1, -1, -1, -1), \quad \mathcal{L}_{\text{int}} = -e\bar{\psi}\gamma^\mu\psi A_\mu - \epsilon e\bar{\psi}\gamma^\mu\psi A'_\mu, \quad e^2 = 4\pi\alpha. \quad (2)$$

The electron is free and initially at rest. We average over its two spins and the two incident photon polarizations, and sum over all final states. We work to leading order in the coupling and set natural units in the derivation. Binding, transport, in-medium mixing, detector response, and later corrections are outside this calculation.

Background formulas used, rather than derived here, are the QED vertex, electron propagator, spin completeness and two-body phase space:

$$V_\gamma^\mu = -ie\gamma^\mu, \quad V_{A'}^\nu = -i\epsilon e\gamma^\nu, \quad S_F(r) = \frac{i(\not{r} + m)}{r^2 - m^2 + i0}, \quad (3)$$

$$\sum_s u_s(p)\bar{u}_s(p) = \not{p} + m, \quad \frac{d\sigma}{dt} = \frac{|\overline{\mathcal{M}}|^2}{16\pi(s - m^2)^2}. \quad (4)$$

Reference 15 is not needed to derive the differential or total cross section. We use its vector-boson result only as an independent numerical check, with its coupling identified as shown below. Park supplies the empirical reactor spectrum and the figure to be tested. No detection reference is needed for this production-only task.

2 Kinematics and the production threshold

In the laboratory define the incident photon energy and the outgoing dark-photon energy by

$$p = (m, \mathbf{0}), \quad k = (E, 0, 0, E), \quad q^0 = x, \quad Q = |\mathbf{q}| = \sqrt{x^2 - M^2}. \quad (5)$$

$$s = (p + k)^2 = m^2 + 2mE, \quad u = (p - q)^2 = m^2 + M^2 - 2mx, \quad (6)$$

$$t = (k - q)^2 = 2m(x - E), \quad s + t + u = 2m^2 + M^2. \quad (7)$$

At threshold both final particles are at rest relative to each other in the centre-of-mass frame. Therefore

$$s \geq (m + M)^2 \implies \boxed{E_{\text{th}} = M + \frac{M^2}{2m}}. \quad (8)$$

The threshold is not merely the dark-photon rest energy. For the plotted masses it is

M [MeV]	E_{th} [MeV]
0.1	0.109785
0.5	0.744619
1	1.978476

Introduce the Källén function and centre-of-mass quantities

$$\lambda(s, m^2, M^2) = [s - (m + M)^2][s - (m - M)^2], \quad (9)$$

$$q_*^0 = \frac{s + M^2 - m^2}{2\sqrt{s}}, \quad q_* = \frac{\sqrt{\lambda(s, m^2, M^2)}}{2\sqrt{s}}, \quad k_* = \frac{s - m^2}{2\sqrt{s}}. \quad (10)$$

Boosting along the beam gives

$$x = \gamma(q_*^0 + \beta q_* \cos \theta_*), \quad \beta = \frac{E}{E + m}, \quad \gamma = \frac{E + m}{\sqrt{s}}, \quad (11)$$

$$\boxed{x_{\pm}(E, M)} = \boxed{\frac{(E + m)(s + M^2 - m^2) \pm E\sqrt{\lambda(s, m^2, M^2)}}{2s}}. \quad (12)$$

The cross section is zero outside these endpoints. The final electron has total energy

$$E_{e'} = m + E - x. \quad (13)$$

3 The two amplitudes and current conservation

The intermediate electron momenta are

$$r_s = p + k, \quad r_u = p - q, \quad a = s - m^2, \quad b = u - m^2. \quad (14)$$

Ignoring an irrelevant common phase, the sum of the two diagrams is

$$\mathcal{M} = \epsilon e^2 \bar{u}(p') \Gamma^{\nu\mu} u(p) \varepsilon_{\mu}(k) \xi_{\nu}^*(q), \quad (15)$$

$$\Gamma^{\nu\mu} = \gamma^{\nu} \frac{\not{r}_s + m}{a} \gamma^{\mu} + \gamma^{\mu} \frac{\not{r}_u + m}{b} \gamma^{\nu}. \quad (16)$$

Both diagrams are required. Contracting with the incident momentum uses

$$\not{k} = (\not{r}_s - m) - (\not{p} - m), \quad (17)$$

$$\not{k} = (\not{p}' - m) - (\not{r}_u - m), \quad (18)$$

$$(\not{r}' - m)(\not{r}' + m) = r^2 - m^2. \quad (19)$$

The on-shell Dirac equations then make the first term give the electron current and the second its negative:

$$k_{\mu} \bar{u}(p') \Gamma^{\nu\mu} u(p) = \bar{u}(p') \gamma^{\nu} u(p) - \bar{u}(p') \gamma^{\nu} u(p) = 0. \quad (20)$$

For the outgoing momentum use

$$\not{q} = \not{q}'_s - \not{p}' = \not{p}' - \not{q}'_u \implies q_\nu \bar{u}(p') \Gamma^{\nu\mu} u(p) = 0. \quad (21)$$

The polarization sums can therefore be contracted as

$$\sum_\lambda \varepsilon_\mu \varepsilon_{\mu'}^* \longrightarrow -g_{\mu\mu'}, \quad (22)$$

$$\sum_{\lambda'} \xi_\nu^* \xi_{\nu'} = -g_{\nu\nu'} + \frac{q_\nu q_{\nu'}}{M^2} \longrightarrow -g_{\nu\nu'}. \quad (23)$$

This does not mean the physical longitudinal polarization is dropped. Its contribution is included by the covariant sum after the Ward identity is used. The independent matrix calculation explicitly sums all three massive-vector polarizations and checks equality. There is no average over final polarizations and no production factor of two-thirds.

4 Evaluating the spin trace

The factor from averaging initial states is

$$\frac{1}{2s_e + 1} \frac{1}{N_{\gamma, \text{pol}}} = \frac{1}{2} \frac{1}{2} = \frac{1}{4}. \quad (24)$$

$$|\overline{\mathcal{M}}|^2 = \epsilon^2 e^4 F, \quad (25)$$

$$F = \frac{1}{4} g_{\mu\mu'} g_{\nu\nu'} \text{Tr} \left[(\not{p}' + m) \Gamma^{\nu\mu} (\not{p} + m) \bar{\Gamma}^{\nu'\mu'} \right], \quad \bar{\Gamma} = \gamma^0 \Gamma^\dagger \gamma^0. \quad (26)$$

Here the denominators remain inside the vertex. To show the trace reduction explicitly, let

$$c = m^2, \quad d = M^2, \quad F = \frac{1}{4} \left(\frac{T_{ss}}{a^2} + \frac{T_{uu}}{b^2} + \frac{T_{su} + T_{us}}{ab} \right). \quad (27)$$

For example, the first term is

$$T_{ss} = \text{Tr} \left[(\not{p}' + m) \gamma^{\nu'} (\not{q}'_s + m) \gamma^\mu (\not{p} + m) \gamma_\mu (\not{q}'_s + m) \gamma_{\nu'} \right]. \quad (28)$$

The Dirac algebra reduces all contractions to scalar products:

$$\text{Tr}(\gamma^\alpha \gamma^\beta) = 4g^{\alpha\beta}, \quad (29)$$

$$\text{Tr}(\gamma^\alpha \gamma^\beta \gamma^\rho \gamma^\sigma) = 4(g^{\alpha\beta} g^{\rho\sigma} - g^{\alpha\rho} g^{\beta\sigma} + g^{\alpha\sigma} g^{\beta\rho}), \quad (30)$$

$$p^2 = c, \quad k^2 = 0, \quad q^2 = d, \quad (31)$$

$$p \cdot k = \frac{a}{2}, \quad p \cdot q = \frac{d-b}{2}, \quad k \cdot q = \frac{a+b}{2}. \quad (32)$$

Longer even traces follow by pairing the first gamma matrix with each subsequent one, with alternating signs. Odd traces vanish. Expanding the four spinor/propagator factors gives

$$T_{ss} = 16cd + 32c^2 + 16ac - 8ab, \quad (33)$$

$$T_{uu} = 16cd + 32c^2 + 16bc - 8ab, \quad (34)$$

$$T_{su} = T_{us} = -8d^2 + 32c^2 + 8bd + 8bc + 8ad + 8ac. \quad (35)$$

Combining the terms gives the compact result

$$F = -2 \left(\frac{a}{b} + \frac{b}{a} \right) + 4(2c + d) \left[\frac{1}{a} + \frac{1}{b} + c \left(\frac{1}{a} + \frac{1}{b} \right)^2 - \frac{d}{ab} \right]. \quad (36)$$

This formula was derived in the new C++ code. It was not imported from a prior implementation or fitted to the figure. The program represents polynomials by their powers of the four invariants. The small dyadic rational coefficients are represented exactly in binary arithmetic. It asserts exact polynomial equality for the compact form, the interference terms, crossing, and the photon limit. Independent complex Dirac matrices provide a separate numerical check.

5 Differential and total cross sections

The laboratory Jacobian and invariant flux give

$$\frac{dt}{dx} = 2m, \quad s - m^2 = 2mE, \quad (37)$$

$$\boxed{\frac{d\sigma_{\text{prod}}}{dx}} = \boxed{\frac{\epsilon^2 e^4}{32\pi m E^2} F(a, b, c, d)}, \quad a = 2mE, \quad b = M^2 - 2mx. \quad (38)$$

For numerical results in centimetres, multiply by the squared conversion constant. The differential cross section then has units of area per energy:

$$\hbar c = 1.973269804 \times 10^{-11} \text{ MeV cm}, \quad \frac{d\sigma}{dx} \left[\frac{\text{cm}^2}{\text{MeV}} \right]. \quad (39)$$

The integration can also be done analytically. With the incident energy fixed, define

$$H = 2c + d, \quad (40)$$

$$G(b) = -\frac{b^2}{a} + 4H \left(\frac{1}{a} + \frac{c}{a^2} \right) b + \left[-2a + 4H \left(1 + \frac{2c-d}{a} \right) \right] \ln \frac{|b|}{a} - \frac{4Hc}{b}, \quad (41)$$

$$\frac{dG}{db} = F, \quad b_{\pm} = M^2 - 2mx_{\pm}, \quad (42)$$

$$\boxed{\sigma_{\text{prod}}} = \boxed{\frac{\epsilon^2 e^4 (\hbar c)^2}{64\pi m^2 E^2} [G(b_-) - G(b_+)]}. \quad (43)$$

The sign follows from the fact that the invariant decreases as the outgoing energy increases. The code verifies the primitive's derivative as an exact polynomial identity and compares the total against direct quadrature and Reference 15.

6 Two independent limits and reference checks

6.1 The photon limit

When the produced vector is massless, the scattering-angle relation and trace become

$$x = \frac{E}{1 + (E/m)(1 - \cos \theta)}, \quad (44)$$

$$F|_{M=0} = 2 \left[\frac{E}{x} + \frac{x}{E} - \sin^2 \theta \right]. \quad (45)$$

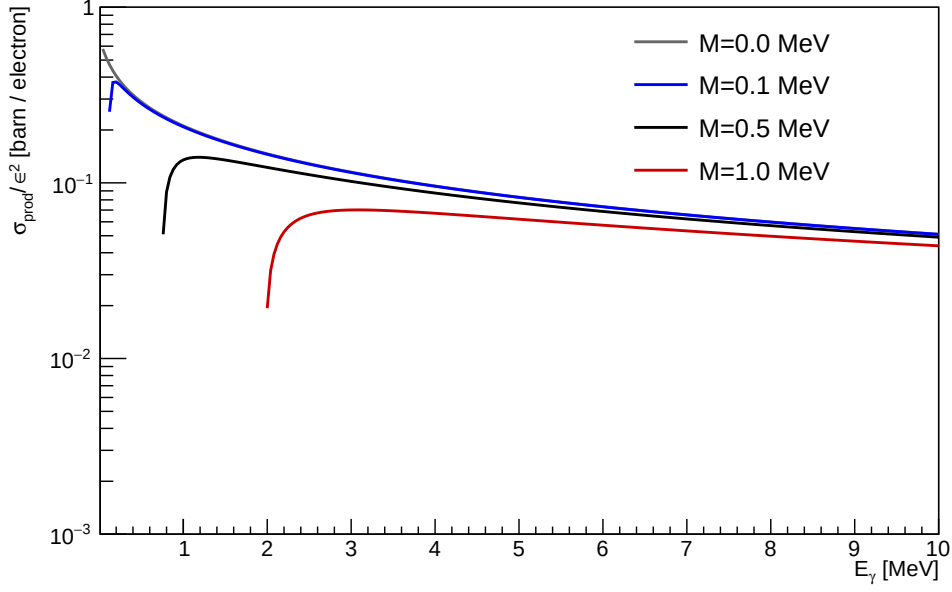


Figure 1: Derived total production cross sections per free electron, with the squared mixing factored out. The massless curve is Klein–Nishina.

For a real final photon, setting the squared mixing coefficient to unity recovers Klein–Nishina. Its total is

$$\sigma_{\text{KN}}(E) = 2\pi r_e^2 \left[\frac{1+y}{y^2} \left(\frac{2(1+y)}{1+2y} - \frac{\ln(1+2y)}{y} \right) + \frac{\ln(1+2y)}{2y} - \frac{1+3y}{(1+2y)^2} \right], \quad (46)$$

$$y = \frac{E}{m}, \quad r_e = \frac{\alpha \hbar c}{m}. \quad (47)$$

6.2 Reference 15

Gondolo and Raffelt’s Appendix uses the following independent total formula for its vector branch, after identifying the coupling:

$$g_{\chi ee} = \epsilon e, \quad \sigma_{15} = \frac{\alpha g_{\chi ee}^2}{8s} \frac{q_*}{k_*} \left[A + B \frac{\sqrt{s}}{q_*} \ln \frac{2q_*^0 E_{e*} + 2q_* k_* - M^2}{2q_*^0 E_{e*} - 2q_* k_* - M^2} \right] (\hbar c)^2, \quad (48)$$

$$E_{e*} = \frac{s + m^2}{2\sqrt{s}}, \quad (49)$$

$$A = 2 + \frac{2(m^2 - M^2)}{s} + \frac{16(M^2 + 2m^2)s}{(s - m^2)^2}, \quad (50)$$

$$B = 2 - \frac{4(M^2 + 2m^2)}{s - m^2} - \frac{4(4m^4 - M^4)}{(s - m^2)^2}. \quad (51)$$

This is a borrowed check, not a step in the derivation. The maximum relative difference on the tested grid is

$$\delta_{15} = 7.06102 \times 10^{-14}, \quad \delta_{\text{KN}} = 7.20535 \times 10^{-14}. \quad (52)$$

These are numerical consistency errors, not experimental uncertainties or global bounds outside the tested grid.

7 From one photon to a reactor spectrum

Park's empirical photon spectrum is a supplied input:

$$S_\gamma(E) = \frac{d\dot{N}_\gamma}{dE} = 0.58 \times 10^{18} \left(\frac{P}{\text{MW}} \right) \exp \left[-\frac{E}{0.91 \text{ MeV}} \right] \text{ s}^{-1} \text{ MeV}^{-1}. \quad (53)$$

It is not derived from fission physics here. Its stated domain starts near the following energy; the text then restricts the study to incident and detected energies above the larger value:

$$E_{\text{spectrum}} \gtrsim 0.2 \text{ MeV}, \quad E_{\text{study}} > 1 \text{ MeV}. \quad (54)$$

We use the stated study cutoff as the baseline and also compute the lower-cutoff alternative, since the figure itself displays dark photons below that study energy. The target is the production spectrum at the reactor, not a detector count. Park's single-interaction source equation is

$$\boxed{\frac{d\dot{N}_{A'}}{dx} = \int_{E_L}^{E_U} S_\gamma(E) \frac{1}{\sigma_{\text{KN}}(E)} \frac{d\sigma_{\text{prod}}(E, x)}{dx} dE.} \quad (55)$$

The numerator and denominator are both per electron. Multiplying both by the same atomic electron count cancels it. This is why the free-electron baseline does not depend on the atomic number. It does not imply that real photon transport is material independent.

The exact incident-energy support follows from the laboratory angle:

$$\cos \theta = \frac{(x - m)E + mx - M^2/2}{EQ}, \quad Q = \sqrt{x^2 - M^2}, \quad (56)$$

$$E_L = \max \left(E_{\text{cut}}, E_{\text{th}}, \frac{mx - M^2/2}{m - x + Q} \right), \quad (57)$$

$$E_U = \begin{cases} \min \left(E_{\text{max}}, \frac{mx - M^2/2}{m - x - Q} \right), & m - x - Q > 0, \\ E_{\text{max}}, & m - x - Q \leq 0. \end{cases} \quad (58)$$

There is no contribution if the positive-angle denominator is nonpositive or the interval is empty. For this mass range the numerator in these bounds is positive. We integrate to the cutoff below and check the larger one for convergence:

$$E_{\text{max}} = 40 \text{ MeV}, \quad E_{\text{max}}^{\text{test}} = 60 \text{ MeV}. \quad (59)$$

The reproduced photon integral at the reference reactor power is

$$P = 1 \text{ mGW}, \quad \int_{1 \text{ MeV}}^{\infty} S_\gamma(E) dE = 1.75883 \times 10^{20} \text{ s}^{-1}, \quad (60)$$

which agrees with the rounded number printed by Park. The plotted choice of mixing is a convention for the leading coefficient, not a controlled prediction at strong mixing.

$$\epsilon = 1 \quad \text{means plotting the coefficient of } \epsilon^2. \quad (61)$$

8 Comparison with Figure 1

We extract horizontal staircase segments from the original PDF vector paths. Each is compared once with the predicted mean over that segment's energy interval. We do not compare both sides of a vertical step as independent data. No normalization is fitted.

$$\bar{S}_i = \frac{1}{x_{i+1} - x_i} \int_{x_i}^{x_{i+1}} \frac{d\dot{N}_{A'}}{dx} dx, \quad r_i = \left| \frac{\bar{S}_i}{S_i^{\text{PDF}}} - 1 \right|. \quad (62)$$

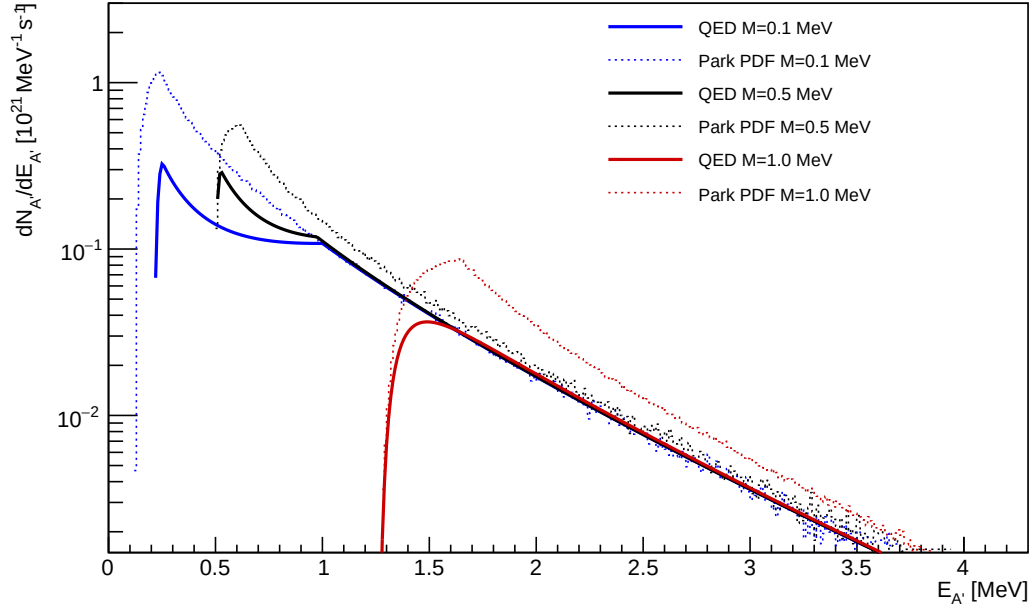


Figure 2: Solid: the new tree-level calculation with Park’s stated source equation. Dotted: coordinates extracted from the supplied Figure 1, not computed data. The axes and rate units match the original.

The central comparison avoids the threshold and the noisy tail. Its selected interval and diagnostic tolerance are

$$1.8 \leq x/\text{MeV} \leq 3.2, \quad \text{median}(r_i) < 0.10. \quad (63)$$

M [MeV]	Bins	Median residual	Maximum
0.1	137	4.15%	21.72%
0.5	136	11.56%	23.69%
1	139	37.85%	55.09%

The tolerance is our stated diagnostic choice, not an uncertainty published by Park. The smallest mass passes this central-interval criterion, while the other masses do not. A central-interval pass cannot certify the complete curve. The lower source cutoff improves the low-energy part of the lightest curve, but cannot change the central interval.

The full Figure 1 is not reproduced. The source-equation calculation and the original figure disagree despite the independent cross-section checks. The source PDF has no original event sample, generator settings, or bin covariance. Those omissions prevent identifying the author’s implementation from the plot alone. Any explanation based on an undocumented generation or normalization choice remains a hypothesis. No such choice has been fitted or substituted into this calculation.

9 When a classical Compton total is not allowed

9.1 Thomson is a low-energy limit

The classical Thomson total is

$$\sigma_T = \frac{8\pi}{3} r_e^2, \quad \sigma_{\text{KN}} = \sigma_T \left[1 - 2\frac{E}{m} + \frac{26}{5} \left(\frac{E}{m} \right)^2 + \dots \right]. \quad (64)$$

It requires incident energies much smaller than the electron mass. With a chosen relative-error criterion, the numerical boundary is

$$\frac{\sigma_T}{\sigma_{\text{KN}}(E)} - 1 < 0.05 \quad \Longleftrightarrow \quad E \lesssim 12.970 \text{ keV} \quad (65)$$

on the low-energy branch. Thomson must not replace Klein–Nishina in the reactor energy range used here.

9.2 Klein–Nishina is one channel, not total material attenuation

The relativistic free-electron formula remains valid for the Compton channel in its stated approximation. A material also permits photoelectric absorption, coherent scattering and pair production. Atomic binding matters near shell energies. Pair production in a nuclear field opens at approximately

$$E_\gamma \geq 2m_e = 1.021997900 \text{ MeV}. \quad (66)$$

This is a threshold, not a statement that pair production dominates immediately. There is no universal energy boundary at which Compton equals the total material cross section. As a concrete check, we compare the following proxy with the NIST uranium total:

$$\left(\frac{\mu}{\rho} \right)_{\text{KN}} = N_A \frac{Z}{A} \sigma_{\text{KN}}, \quad Z = 92, \quad A = 238.02891 \text{ g mol}^{-1}. \quad (67)$$

E [MeV]	$(\mu/\rho)_{\text{NIST}}$ [cm ² /g]	KN/NIST	$\sigma_T/\sigma_{\text{KN}}$
0.2	1.29800	0.073	1.64
1	0.07896	0.623	3.15
2	0.04878	0.698	4.55
3	0.04447	0.602	5.78
5	0.04463	0.432	8.03
8	0.04879	0.286	11.11

Over the tabulated sample, the free-electron Compton proxy never reaches the precision criterion below. We retain Park’s denominator for the figure test, but do not endorse it as precision total attenuation. The NIST comparison is a diagnostic; it does not modify the source spectrum.

$$0.1 \leq E/\text{MeV} \leq 10, \quad \left| \frac{(\mu/\rho)_{\text{KN}}}{(\mu/\rho)_{\text{NIST}}} - 1 \right| < 0.10. \quad (68)$$

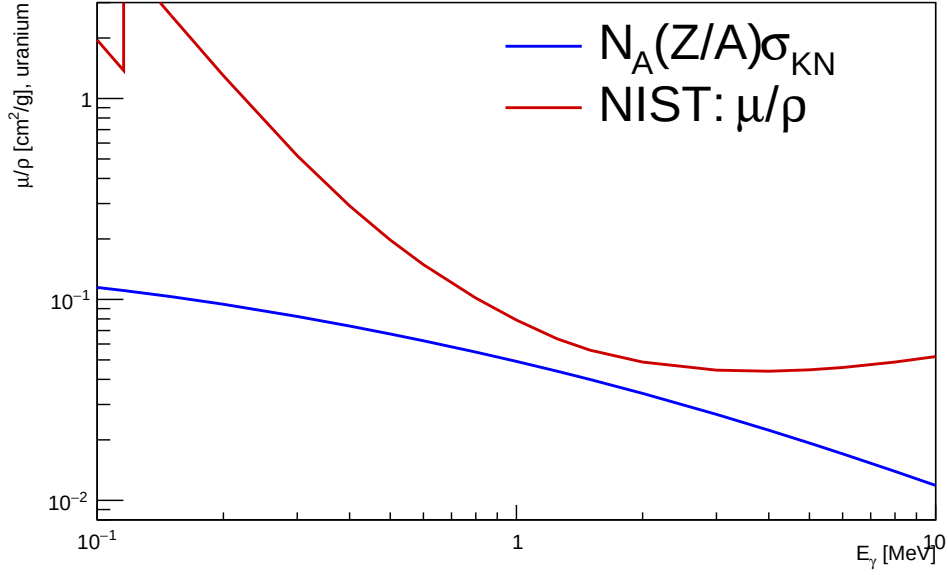


Figure 3: Free-electron Compton proxy versus the total uranium attenuation from NIST. The material dependence is missing from a Compton-only denominator.

10 Implementation, checks and limits of the result

All new calculation, extraction and reporting utilities are C++ using ROOT. The executable was compiled with the requested environment. LaTeX produces the lecture PDFs. There is no Python component. The code is separated into named functions for the trace, kinematics, cross sections, source convolution, PDF extraction, comparison and reports.

The source configuration exposes thermal power, spectrum normalization and slope, incident-energy bounds, and quadrature order. A different measured source can replace the photon-spectrum function. Background, exposure, detector mass, region of interest and target composition belong to a subsequent detection calculation; they do not change the production cross section. For a later electron target the bookkeeping factor would be

$$N_e = m_{\text{det}}[\text{g}] N_A \frac{Z}{A}. \quad (69)$$

No detection model or acceptance has been invented to make those parameters enter this production-only result.

Checks actually run: exact polynomial identities; Clifford algebra; independent complex-matrix traces and physical polarizations; both Ward identities on an on-shell grid; analytic totals versus quadrature and Reference 15; Klein–Nishina; source-node doubling and upper-tail extension; the printed photon integral; and direct PDF-bin comparisons. The tested source-node difference is

$$\delta_{\text{nodes}} = 1.80078 \times 10^{-13}. \quad (70)$$

Reversing the source integration order, by first choosing the incident photon and then integrating over the centre-of-mass-derived outgoing support, also agrees in the tested energy bins. This checks the source-convolution support independently of the outgoing-energy-first calculation. Not checked: the empirical reactor spectrum against its original measurement, reactor transport, bound-electron effects, loop corrections, the author’s unpublished generator and event statistics. The matrix Ward tests sample physical points; the report also provides their analytic proof.

Numerical tests do not establish accuracy outside the tested domain. The full published Figure 1 remains unreproduced.

References

- [1] H. K. Park, *Detecting Dark Photons with Reactor Neutrino Experiments*, Phys. Rev. Lett. 119, 081801 (2017). [doi:10.1103/PhysRevLett.119.081801](https://doi.org/10.1103/PhysRevLett.119.081801). Local source: papers/park_production_v2/park2017.pdf.
- [2] P. Gondolo and G. G. Raffelt, *Solar neutrino limit on axions and keV-mass bosons*, Phys. Rev. D 79, 107301 (2009), Appendix. [arXiv:0807.2926](https://arxiv.org/abs/0807.2926). Local source: papers/park_production_v2/reference15.pdf.
- [3] NIST, *X-Ray Mass Attenuation Coefficients: Uranium*. <https://physics.nist.gov/PhysRefData/XrayMassCoef/ElemTab/z92.html>. Downloaded source retained locally.
- [4] ROOT, *Numerical integration reference, version 6.36*. https://root.cern.ch/doc/v636/classROOT_1_1Math_1_1GaussLegendreIntegrator.html.